Nathaniel Kessel

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EDUCATION

University of Cincinnati, Cincinnati, Ohio — Expected Graduation: May 2028
Bachelor of Science, Computer Engineering

Miami Valley Career Technology Center (MVCTC), Clayton, Ohio — August 2021 – June 2023

- Built foundation for troubleshooting and maintenance practices using multimeters, calipers, safety wire tooling, soldering, and other equipment
- Practiced rigorous Aviation Maintenance Procedures as prescribed in official FAA documentation, ensuring strict compliance with industry regulations and safety standards

PROJECTS

ESP32 Desk Clock — September 2025 – Present
Personal Project

- Developing a compact ESP32-S3 desk clock featuring an SPI-driven LED display, real-time clock (RTC), and NTP network time synchronization
- Prototyped circuit design and firmware in Arduino C++, with plans to add weather, calendar, and local data integration
- Troubleshot PCB revisions utilizing digital multimeters and oscilloscopes to verify signal integrity and board design

Engineering Design Thinking 2 — January 2024 – May 2024
University of Cincinnati, Cincinnati, Ohio

- Led a team of four individuals to create a robot capable of navigation to a specified location utilizing ultrasonic and gyroscopic sensors coded in Python
- Managed team cooperation and development to stay on task for the final tests, earning a score above average and staying on track throughout the semester

Engineering Design Thinking 1 — April 2023 – December 2023
University of Cincinnati, Cincinnati, Ohio

- Collaborated with a team of four individuals to create a robot capable of movement, lifting, and general navigation utilizing LabVIEW
- Led Code of Conduct development incorporating multiple team building exercises to bring project completion in a timely manner to achieve an 83%, surpassing average

EXPERIENCE

Closing Expert, **Target**, Milford, Ohio — September 2023 – Present

- Balanced an average of 26 hours a week with full-time academic courseload
- Trained and guided employees on key WMS store metrics and utilizing them for improved end-of-day organization and performance metrics
- Improved fill percentage from an average of 70% to consistently over 85% during the 2024 season
- Managed high-volume customer interactions, resolving escalations and educating guests on product offerings

SKILLS

Programming: C++, C, LabVIEW, MATLAB, Python, VBA, HTML/CSS, Arduino, Verilog
Hardware & CAD: KiCad 9.0, Fusion360, AutoCAD, LTspice, Multisim, Vivado
Productivity: Microsoft Excel, Microsoft Office Suite, Git

AVAILABLE TO CO-OP FALL 2026